

Business Intelligence

LEARNING OUTCOMES

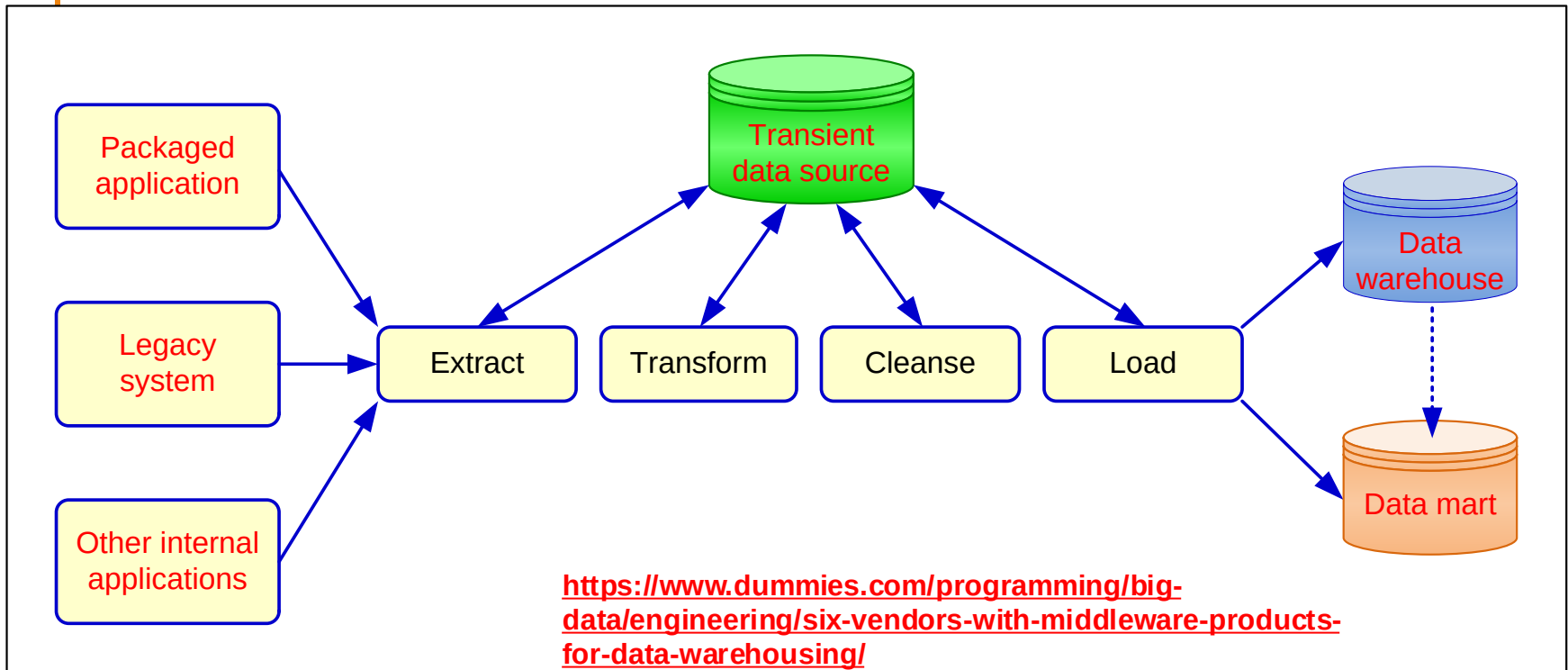
1. Explain the relationship between Business Intelligence and a Data Warehouse
2. Data multi-dimensionality
3. Steps in BI

Data Warehouse Fundamentals

- **Data warehouse** – a logical collection of information – gathered from many different operational databases – that supports business analysis activities and decision-making tasks



Data Integration and the Extraction, Transformation, and Load (ETL) Process



Multidimensional Analysis and Business Intelligence

- Databases contain information in a series of two-dimensional tables
- *In a Business Intelligence, data is multidimensional:*
 - It contains layers of columns and rows
 - Dimension – a particular attribute of information



Representation of Data in DW

◦ Dimensional Modeling

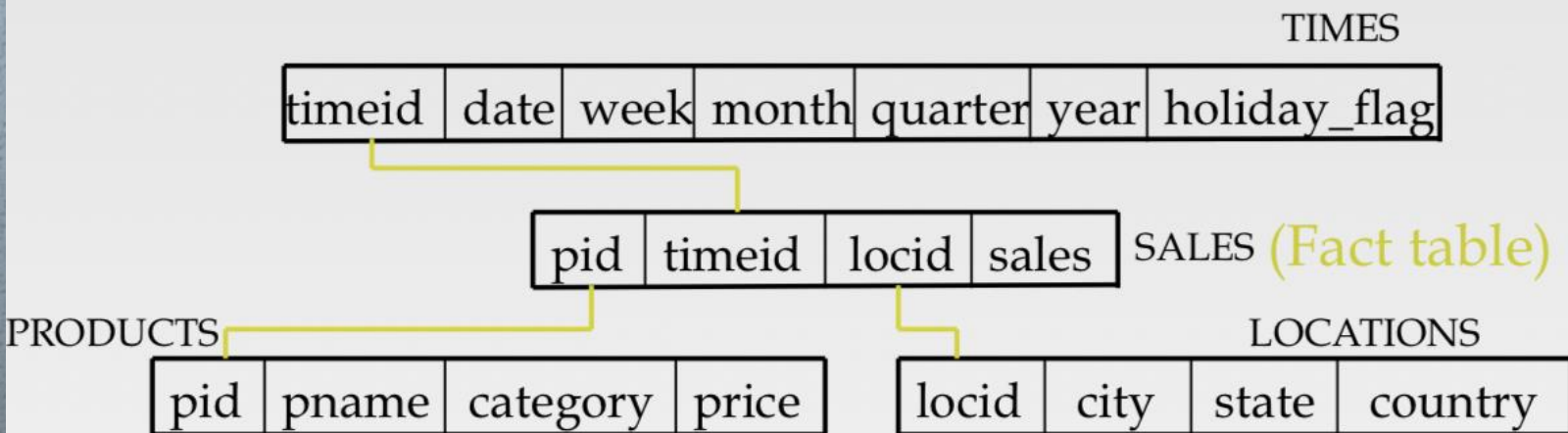
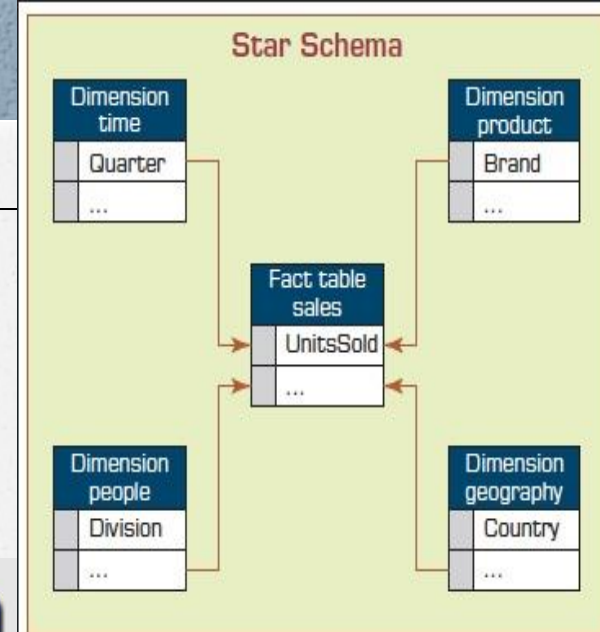
- A retrieval-based system that supports high-volume query access

◦ Star schema (Logical structure)

- The most commonly used and the simplest style of dimensional modeling
- Contain a **fact table** surrounded by and connected to several **dimension tables**

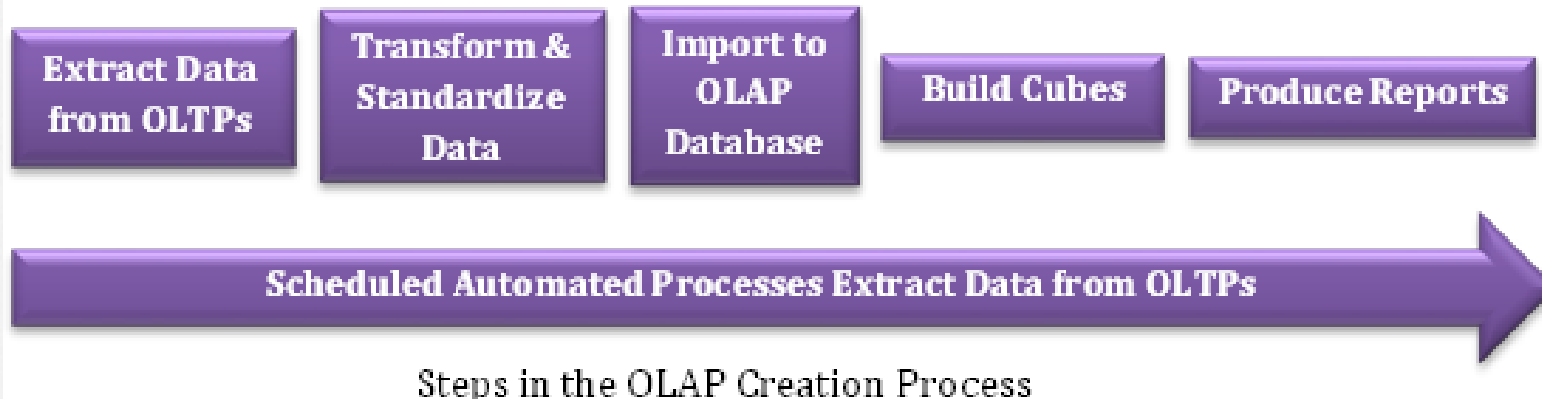
Star Schema versus Snowflake Schema

Star Schema Design



- Fact table is large, updates are frequent; dimension tables are small, updates are rare.
- This kind of schema is very common in OLAP applications, and is called a **star schema**; computing the join of all these relations is called a **star join**.

OLAP Creation Process

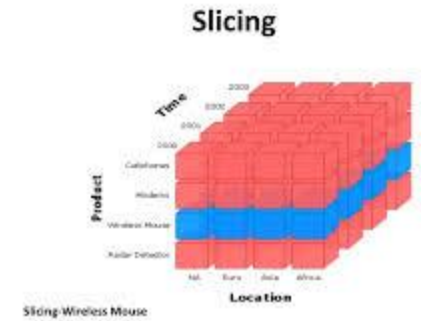
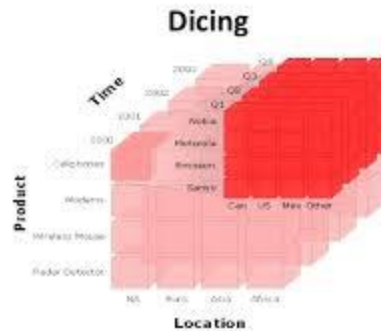


- o 2 usual ways to get action on OLAP:
 - o **ROLAP:** Relational OLAP, use SQL to query directly on relational tables.
 - o **MOLAP:** Multidimensional OLAP, data **aggregated** as flat file called OLAP cube for **fast slicing, dicing, roll-up, and drill-down** operations using MDX (Multidimensional Expression).
 - o **In Excel:**
 - o Pivot table ~ MOLAP (deal with flat table)
 - o Power Pivot ~ ROLAP (deal with relational tables)



OLAP Operations

- **Slice** - a subset of a multidimensional array
 - Slice on Year 2022 and Region North
- **Dice** - a slice on more than two dimensions
 - Dice to see only Year 2022-2023, Products (A, B), Region(North, South)
- **Drill Down/(Roll) Up** - navigating among levels of data ranging from the most summarized (up) to the most detailed (down)
 - Year → Drill down to Year/Quarter → Month
 - Year/Quarter → Roll up to Year
- **Pivot** - used to change the dimensional orientation of a report or an ad hoc query-page display
 - Region on X, Time on Y, Product on Z-axis → Time, Product, Region



BUSINESS INTELLIGENCE

- o *Business intelligence (BI)* – applications and technologies used to gather, provide access to, and analyze data and information to support decision-making efforts
- o Paraphrase the challenges in business and challenges in war
 - o Collecting information
 - o Discerning patterns and meaning in the information
 - o Responding to the resultant information

THE SOLUTION: BUSINESS INTELLIGENCE

Question

Why are sales below target?

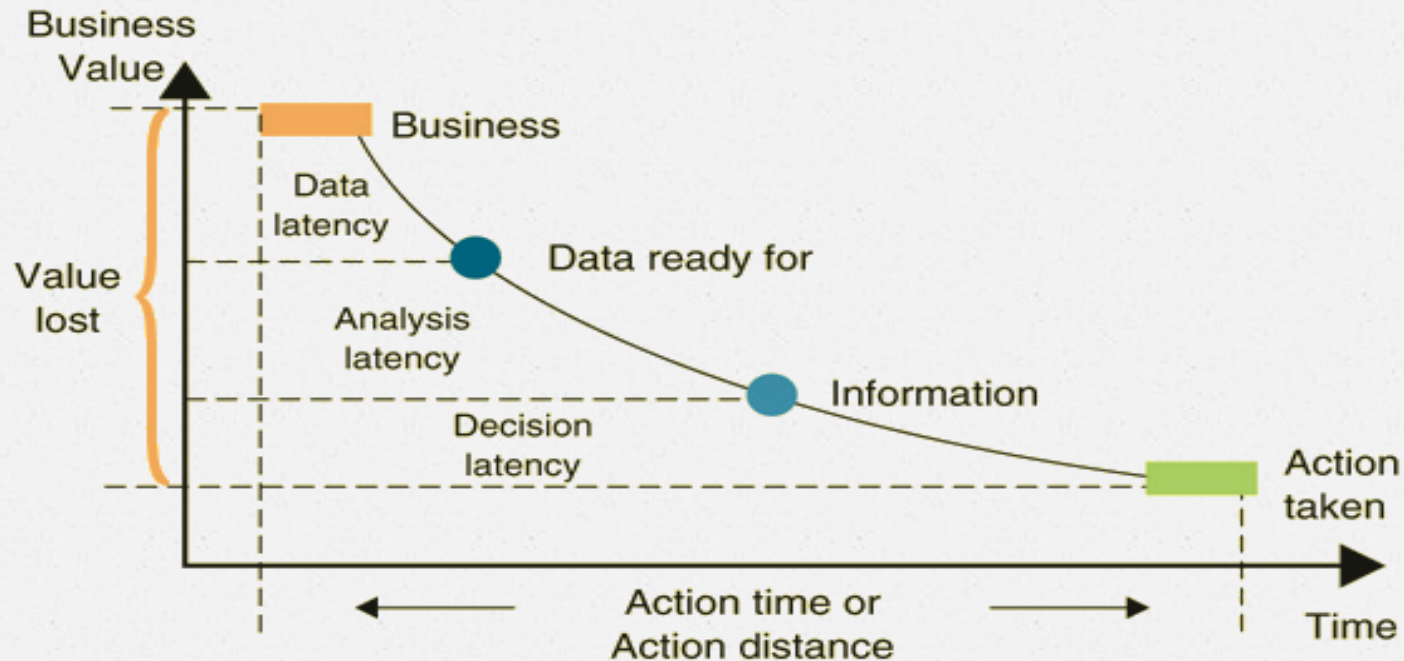
Answer

Because we sold less
in the Western region.

BI Can Answer Tough Questions

BI's Operational Value

- Richard Hackathorn's graph demonstrating the value of operational BI



Objective learning review

- o DW is subjected-oriented DB specially designed to support decision making without disrupting daily operations.
- o OLAP and BI are tools equipped to DW for analytics.